SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

YUGITHA B
192424102RD

QUESTIONS 10 / 10 completed

Single Linked List

SINGLE LINKED LIST

"A college maintains a list of student roll numbers in the order they are registered. New students may join at different stages of the registration process. Develop a C program using a singly linked list to store the roll numbers and perform insertion operations at the beginning, end, and specified position. Display the updated list after each insertion operation.

Test Case 1

Input
Initial List: Empty
Insert 10 at beginning
Insert 20 at end
Insert 15 at position 2
Output
10 → 15 → 20 → NULL"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* next;
};

struct Node* createNode(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct N
    if (!newNode) {
        printf("Memory allocation failed!\n");
        exit(1);
    }
    newNode->data = value;
    newNode->next = NULL;
    return newNode;
}

void insertBeginning(struct Node** head, int value) {
    struct Node* newNode = createNode(value);
```

OUTPUT

```
Enter number of insert operations:
Operation 1
1. Insert at beginning
2. Insert at end
3. Insert at position
Choose option (1/2/3): Invalid choice. Skipping operation.

Operation 2
1. Insert at beginning
2. Insert at end
```

INPUT

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